

Madhav Venkatachalam

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EDUCATION

Cornell University

Ithaca, NY

B.S. in Applied and Engineering Physics, Minor in Mathematics

May 2029

- **GPA:** 4.03/4.00
- **Relevant Coursework:** Calculus 3, Differential Equations, Linear Algebra, Real Analysis, Number Theory, Honors Mechanics and Special Relativity, Honors Electricity and Magnetism, Honors Waves and Oscillations, Experimental Physics 1&2, Lasers and Photonics, Quantum Information and Hardware

EXPERIENCE

Undergraduate Researcher

September 2025 – Present

Fuchs Group (Cornell University)

Ithaca, NY

- Simulated high-throughput microwave packages for precise superconducting device measurement using COMSOL and Ansys; Designed a modified package using Autodesk Fusion to minimize parasitic modes
- Tested optical modes in SiC waveguides and ring resonators using a tunable cavity coupling to spin defects
- Designed and fabricated a novel PCB and backing plate to better house multiple 10mm superconducting chips

Teaching Assistant

July 2026 – August 2026

Massachusetts Institute of Technology

Remote

- Taught MIT Beaver Works Summer Institute's Quantum Software course alongside other teaching staff
- Taught quantum algorithms such as Deutsch-Jozsa, Bernstein-Vazirani, Simon's, Grover's, and Shor's
- Guided final projects (e.g. Variational Quantum Eigensolver and QAOA for MaxCut) in Qiskit and Q#

Problem Design Engineer

March 2026 – Present

Idlexperts/Idler (YC S25)

Remote

- Evaluated agent-generated failure analyses for Idler's RL training pipeline, and built environments for testing
- Reviewed production-level code changes across repos to evaluate problem clarity and fairness for training datasets
- Produced concise technical justification and flag coherence issues such as ambiguous requirements, inconsistent artifacts, or missing implementation context

Undergraduate Researcher

December 2025 – May 2026

Katz Lab (Cornell University)

Ithaca, NY

- Designed low-pass RLC filters for coherent quantum entanglement of a Barium trapped ion quantum computer
- Built a temperature monitoring system for a laser-cooled optical ion trap. Engineered optical housing for the trap

EXTRACURRICULARS & PROJECTS

Cornell Hyperloop | Magentics Subteam

September 2026 – Present

- Utilized Ansys Maxwell to design and optimize a linear induction motor (LIM), doubling thrust output compared to prior competition designs
- Directed magnetics development and coordinated cross-functional subsystem design, integration, and implementation across Electrical, Mechanical, and Software subteams
- Designed, simulated, and built a novel EMS magnetic levitation system for the competition hyperloop pod and track moving away from the previously used EDS system

Quantum Computing Association | Theory and Decoding Subteam

September 2025 – Present

- Theory Subteam: Worked on developing Dynamic Decoupling codes. Worked on developing quantum low density parity check codes through Z-X calculus
- Decoding Subteam: Worked on optimizing Union Find algorithm and Blossom algorithm for MWPM

Cornell Venture Capital | Analyst

February 2026 – Present

- Worked with Insight Partners and Foundation Capital to deliver analysis and early-stage investment opportunities
- Focused on market analysis of the Hardware layer of the physical intelligence and robotics industries

SKILLS AND INTERESTS

Technical: Java, C/C++, Qiskit, Q#, Python, MATLAB, Stim, COMSOL, LTSpice, Ansys, Fusion, AutoCAD

Languages: English, Tamil, Hindi, French

Interests: Quantum and Photonic Computing, Experimental Condensed Matter Physics, Vinyl Records